Octopus: Alleviating Hallucination via Dynamic Contrastive Decoding

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Abstract

Large Vision-Language Models (LVLMs) have achieved remarkable progress in multimodal understanding, yet they continue to suffer from hallucinations—generating content that contradicts the visual input. Existing Contrastive Decoding (CD) methods apply a single, fixed strategy to all inputs and all generation steps, overlooking the fact that hallucination causes are hybrid: different samples and different decoding steps are driven by distinct factors such as language prior bias, visual information loss, and attention misalignment. In this paper, we propose ***Octopus****, a plug-and-play framework that performs dynamic, per-token selection among multiple CD strategies. Octopus introduces a lightweight transformer module with a learnable decision token (*eye*) that attends to the LVLM's hidden states at each decoding step and selects the most suitable CD "tentacle"—VCD, M3ID, AVISC, or a null action—from a candidate pool. The LVLM's weights remain fully frozen; only the Octopus module is trained using Direct Preference Optimization (DPO) on preference pairs constructed by sampling action sequences and ranking with hallucination criteria. Extensive experiments demonstrate that Octopus achieves state-of-the-art hallucination reduction on both generative benchmarks (MMHalBench, Object-HalBench, AMBER) and discriminative benchmarks (POPE-MSCOCO), generalizing across LLaVA-1.5 and InstructBLIP. Notably, Octopus reduces the AMBER CHAIR score by 40% (8.0→4.8) and improves AMBER discrimination F1 by 11.6 points (71.1→82.7) on LLaVA-1.5-7B.*

1. Introduction

Large Vision-Language Models (LVLMs) [1, 2, 7, 18, 19, 40] have made impressive strides in tasks ranging from image captioning to visual question answering. However, they remain prone to *hallucinations*: generating plausible but factually incorrect descriptions of objects, attributes, or relations that do not appear in the image. This critical deficiency poses serious obstacles for real-world deployment.

Prior work on hallucination mitigation falls into two broad camps. *Training-based* approaches [11, 26, 28, 36–39] fine-tune the LVLM itself using specially curated datasets or reinforcement learning from human feedback (RLHF), but incur significant computational cost and risk degrading general capabilities. *Training-free* decoding strategies [8, 9, 14, 20, 33, 35] apply contrastive decoding (CD) at inference time, contrasting original outputs with those from a disturbed input to suppress hallucinations without modifying model weights.

Despite their appeal, existing CD methods share a key limitation: they apply a *single, uniform* strategy across all inputs and all generation steps. As illustrated in Figure 1, each CD strategy targets a specific hallucination cause: Visual Contrastive Decoding (VCD) [14] counters statistical language priors; Multi-Modal Mutual-Information Decoding (M3ID) [8] recovers fading visual grounding during long generation; Attentional Vision Calibration (AVISC) [35] reduces attention bias toward irrelevant tokens. Our empirical analysis (Figure 2) reveals that each strategy is effective only on a subset of samples—only $\sim$10% of hallucinated cases are resolved by all three methods simultaneously. Moreover, even within a single generated sequence, the dominant hallucination cause shifts from one token to the next, making static strategy selection inherently suboptimal.

These findings motivate a fundamentally different approach: *dynamic, token-level CD strategy selection* tailored to the current multimodal context. We propose **Octopus**, a lightweight plug-and-play module that acts as an intelligent orchestrator—like an octopus coordinating its tentacles—selecting the most appropriate CD strategy at every decoding step. Concretely, Octopus adds a small transformer encoder block with a learnable *decision token* (the "eye") that attends to the LVLM's hidden states and outputs a distribution over the CD strategy pool: {VCD, M3ID, AVISC, Null}. The selected strategy then constructs the contrastive input and adjusts logits accordingly. The LVLM's weights are kept frozen throughout, and Octopus is trained end-to-end using Direct Preference Optimization (DPO) [24] on preference pairs built by sampling multiple action sequences and scoring them with hallucination met-

rics.

Our contributions are:

- We empirically demonstrate that hallucination causes are *hybrid* at both the sample and token levels; single CD strategies are effective only on specific subsets of hallucinated cases.
- We propose **Octopus**, a dynamic per-token CD controller consisting of a decision token and a compact transformer encoder that selects among multiple CD strategies based on contextual hidden states.
- We introduce a DPO-based training paradigm for Octopus that learns optimal action workflows without LVLM retraining or explicit reward models.
- Octopus achieves state-of-the-art performance across generative and discriminative hallucination benchmarks, with strong generalization across model families and easy extensibility to new CD strategies.

2. Related Work

Large Vision-Language Models. LVLMs connect pre-trained visual encoders (e.g., CLIP [23]) with large language model backbones [6, 21, 29] through cross-modal alignment modules. Prominent examples include LLaVA [19], LLaVA-1.5 [18], MiniGPT-4 [40], InstructBLIP [7], and Qwen-VL [2]. While highly capable, these models suffer from hallucinations [16, 25, 32].

Training-Based Hallucination Mitigation. One line of work fine-tunes LVLMs using data augmentation or preference learning to reduce hallucinations. HACL [11] introduces hallucination-augmented contrastive learning; POVID [39] and HA-DPO [38] apply DPO with hallucination-aware preference pairs; EOS [37] targets end-of-sequence decisions; HALVA [26] uses phrase-level contrastive tuning; RLHF-V [36] leverages fine-grained human feedback. These methods require LVLM retraining and may degrade general capabilities.

Contrastive Decoding for Hallucination. Training-free CD approaches contrast the output distribution of the original LVLM with that of a disturbed variant. VCD [14] adds noise to the visual input to amplify language priors and subtract them. M3ID [8] amplifies the mutual information between the visual prompt and generated tokens to maintain visual grounding. AVISC [35] identifies attention-misaligned "blind" tokens in the image and applies CD to calibrate their influence. ICD [33] perturbs text instructions rather than visual inputs. OPERA [9] applies an over-trust penalty in beam search. HALC [4] combines local grounding and global beam search. SID [10] performs self-introspective decoding based on layer-wise importance. ConVis [22] uses text-to-image reconstruction for visual contrast. LCD [20] adjusts outputs based on language model confidence. DoLa [5] contrasts logits across transformer layers to improve factuality.

Despite their effectiveness on specific hallucination types, all existing CD methods apply a fixed strategy uniformly. Our work fundamentally differs by learning to dynamically select among multiple strategies per decoding step.

Preference Optimization. DPO [24] reformulates RLHF as a simple classification loss, avoiding explicit reward models. mDPO [31] extends DPO to multimodal settings. Silkie [15] applies DPO with AI-annotated visual feedback. We leverage DPO to train the Octopus strategy selector, using hallucination metrics as implicit rewards.

3. Preliminary: Contrastive Decoding

We briefly review the three CD strategies used as Octopus tentacles.

Visual Contrastive Decoding (VCD) [14]. VCD mitigates object hallucinations caused by statistical language priors. At each decoding step t, a distorted image $\tilde{v}$ is created by adding Gaussian diffusion noise to the original visual tokens v. The adjusted logit distribution is:

$$p_{\text{VCD}}(y_t|\mathbf{x}) \propto p(y_t|v, \mathbf{x}_{<t}) - \alpha \cdot p(y_t|\tilde{v}, \mathbf{x}_{<t}), \quad (1)$$

where $\alpha > 0$ is a contrastive weight and $\mathbf{x}_{<t}$ denotes the previously generated tokens. By subtracting the distribution induced by distorted visual input, VCD suppresses tokens driven primarily by language priors.

Multi-Modal Mutual-Information Decoding (M3ID) [8]. M3ID addresses the phenomenon where visual grounding fades during longer text generation. It amplifies the mutual information between the visual prompt and the next token by scaling the conditional visual probability:

$$p_{\text{M3ID}}(y_t|\mathbf{x}) \propto p(y_t|\mathbf{x}_{<t})^{1-\beta} \cdot p(y_t|v, \mathbf{x}_{<t})^{\beta}, \quad (2)$$

where β controls the degree of visual amplification. This ensures that tokens with higher mutual information with the image are favored over those predicted purely from language context.

Attentional Vision Calibration (AVISC) [35]. AVISC targets hallucinations arising from attention bias toward visually irrelevant image tokens ("blind tokens"). It identifies blind tokens through layer-wise attention analysis, masks them in a distorted input v_{mask}, and applies CD:

$$p_{\text{AVISC}}(y_t|\mathbf{x}) \propto p(y_t|v, \mathbf{x}_{<t}) - \gamma \cdot p(y_t|v_{\text{mask}}, \mathbf{x}_{<t}), \quad (3)$$

where γ modulates the contrastive strength.

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Null Action. Octopus also maintains a null action, corresponding to standard greedy decoding $p(y_t|v, \mathbf{x}_{<t})$ without any contrastive adjustment. This is appropriate when the current token is unlikely to be hallucinated.

4. Octopus: Dynamic Contrastive Decoding

4.1. Overview

Figure 4 illustrates the Octopus framework. At each autoregressive decoding step t, the LVLM produces hidden states $\mathbf{h}_t \in \mathbb{R}^{L\times d}$ from its transformer layers, where L is the sequence length and d is the hidden dimension. Rather than applying a fixed CD strategy, Octopus uses a lightweight module to examine $\mathbf{h}_t$ and select the most appropriate action $a_t \in \{\text{VCD}, \text{M3ID}, \text{AVISC}, \text{Null}\}$.

4.2. Decision Token Architecture

The Octopus module consists of three components: a learnable *decision token* (cls token), a small transformer encoder, and an MLP classifier.

Decision Token. We introduce a single learnable parameter $\mathbf{e} \in \mathbb{R}^{1\times d}$ (the "eye"), analogous to the `[CLS]` token in BERT [13]. This token is prepended to the LVLM's hidden state sequence at each step:

$$\mathbf{H}_t = \text{concat}(\mathbf{e}, \mathbf{h}_t) \in \mathbb{R}^{(L+1)\times d}. \tag{4}$$

Transformer Encoder. $\mathbf{H}_t$ is passed through a small transformer encoder [30] with 2 layers and 2 attention heads, operating in float32 precision for training stability:

$$\mathbf{Z}_t = \text{TransformerEncoder}(\mathbf{H}_t) \in \mathbb{R}^{(L+1)\times d}. \tag{5}$$

The encoder captures contextual relationships between the decision token and the full hidden state sequence, allowing it to attend to both visual and textual representations.

MLP Classifier. The output corresponding to the decision token position, $\mathbf{z}_t^{[0]} \in \mathbb{R}^d$, is fed into a two-layer MLP to produce action logits:

$$\mathbf{o}_t = \text{MLP}(\mathbf{z}_t^{[0]}) = W_2 \cdot \text{LeakyReLU}(W_1 \cdot \mathbf{z}_t^{[0]}) \in \mathbb{R}^4, \tag{6}$$

where $W_1 \in \mathbb{R}^{d/4\times d}$ and $W_2 \in \mathbb{R}^{4\times d/4}$ are initialized with Kaiming uniform initialization. The selected action is $a_t = \arg\max(\mathbf{o}_t)$.

4.3. Strategy Execution (Tentacles)

Given the selected action a_t, the corresponding CD strategy constructs the appropriate contrastive input (if any) and adjusts the token logits before the softmax step. The four strategies are:

- a_t = VCD: Construct $\tilde{v}$ by adding diffusion noise [14] at noise step 500 to the visual tokens and apply Eq. (1).
- a_t = M3ID: Amplify visual conditioning as in Eq. (2).
- a_t = AVISC: Identify blind tokens via layer-wise attention analysis, construct masked visual input v_{mask}, and apply Eq. (3).
- a_t = Null: No contrastive adjustment; use the original distribution.

Crucially, the LVLM's weights remain completely frozen throughout both training and inference. Only the parameters $\{\mathbf{e}, \text{TransformerEncoder}, \text{MLP}\}$ are trained.

4.4. DPO-Based Training

Preference Pair Construction. To train Octopus without reward models, we construct preference pairs by sampling multiple action sequences for each training image-question pair. For a training sample (v, q), we run Octopus with stochastic action sampling to produce K different action sequences $\{a_{1:T}^{(k)}\}_{k=1}^{K}$ and their corresponding generated responses $\{y^{(k)}\}_{k=1}^{K}$. Each response is scored using the CHAIR metric [25] (for generative tasks) or task accuracy (for discriminative tasks). The highest-scoring response and its action sequence serve as the *chosen* pair $(y_w, a_{1:T}^w)$, and the lowest-scoring response serves as the *rejected* pair $(y_l, a_{1:T}^l)$.

DPO Objective. We apply DPO [24] directly to the action selection distribution. Let $\pi_\theta(a_{1:T}|\mathbf{h}_{1:T})$ denote the Octopus policy. The DPO loss is:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}\Big[\log\sigma\Big(\beta_{\text{dpo}}\log\frac{\pi_\theta(a_{1:T}^w|\mathbf{h}_{1:T})}{\pi_{\text{ref}}(a_{1:T}^w|\mathbf{h}_{1:T})} - \beta_{\text{dpo}}\log\frac{\pi_\theta(a_{1:T}^l|\mathbf{h}_{1:T})}{\pi_{\text{ref}}(a_{1:T}^l|\mathbf{h}_{1:T})}\Big)\Big], \tag{7}$$

where π_{ref} is the reference policy (uniform over actions) and β_{dpo} is the regularization coefficient. In practice, since the reference policy is uniform (no initial preference), the DPO loss reduces to contrasting the log-probabilities of chosen versus rejected action sequences under the learned policy.

This training paradigm avoids explicit reward model training, LVLM retraining, or reinforcement learning with environment interaction—the Octopus module learns purely from pairwise preferences derived from hallucination metrics.

4.5. Extensibility

The Octopus framework is inherently extensible: new CD strategies can be added as additional tentacles by simply expanding the MLP output dimension from 4 to $4 + N_{\text{new}}$ and retraining Octopus. The LVLM itself never needs to be modified. This plug-and-play design makes Octopus compatible with any autoregressive LVLM that exposes hidden states during generation.

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5. Experiments

5.1. Setup

Base Models. We evaluate Octopus on two representative LVLMs:

- **LLaVA-1.5-7B** [18]: A strong LVLM using CLIP-ViT-L-336px with an MLP connector and Vicuna-7B.
- **InstructBLIP-7B** [7]: A BLIP-2-based model with an instruction-aware Q-Former and Vicuna-7B.

Benchmarks. We evaluate on five benchmarks spanning both generative and discriminative hallucination:

- **MMHalBench** [28]: Measures both hallucination rate (HalRate) and overall quality score (1–6).
- **Object-HalBench** [25]: Reports CHAIR_s (sentence-level) and CHAIR_i (instance-level) hallucination rates in captions.
- **AMBER** [32]: An LLM-free benchmark evaluating both generative (CHAIR, Coverage, HalRate, Cognition) and discriminative (Accuracy, F1) hallucination.
- **POPE-MSCOCO** [16]: A polling-based benchmark measuring object hallucination in yes/no questions under random, popular, and adversarial settings.

Implementation Details. Training uses AMBER generative data with MSCOCO [17] images. We sample $K = 8$ action sequences per training instance and use CHAIR to construct preference pairs. The Octopus transformer encoder has 2 layers and 2 attention heads ($d = 4096$ matching LLaVA-1.5's hidden dimension). The MLP uses LeakyReLU activation. We train with the Adam optimizer [12] for 5 epochs. Inference adds negligible overhead as it requires only one forward pass through the small Octopus module per decoding step. We reproduce VCD, M3ID, and AVISC using official implementations († indicates our reproductions).

5.2. Main Results: Generative Tasks

Table 1 compares Octopus against state-of-the-art methods on generative hallucination benchmarks. We group results into referenced (fine-tuning-based, not directly comparable) and comparable (CD-based) methods.

On **LLaVA-1.5-7B**, Octopus achieves:

- **MMHalBench**: Score improves from 1.59 to **2.61** (+64%), HalRate drops from 0.72 to **0.50**, surpassing all CD baselines (VCD: 1.96, M3ID: 2.14, AVISC: 2.19).
- **Object-HalBench**: CHAIR_s improves from 25.0 to **20.8** (-16.8%), CHAIR_i from 9.2 to **6.6** (-28.3%).
- **AMBER (Generative)**: CHAIR drops from 8.0 to **4.8** (-40%), Coverage increases from 44.5 to **49.2**, HalRate from 31.0 to **23.4**, and Cognition from 2.2 to **1.2**.

Notably, Octopus outperforms even the fine-tuning-based methods (HACL: 2.13, POVID: 2.08, EOS: 2.03, HA-DPO: 1.97, HALVA: 2.25) on MMHalBench without modifying LVLM weights, demonstrating the power of dynamic strategy selection.

On **InstructBLIP-7B**, Octopus generalizes strongly: MMHalBench score improves from 1.84 to **2.31**, HalRate from 0.64 to **0.49**, and AMBER CHAIR from 8.4 to **6.1**. This cross-model generalization validates the model-agnostic design of Octopus.

5.3. Main Results: Discriminative Tasks

Table 2 presents results on discriminative benchmarks.

On **LLaVA-1.5-7B**, Octopus achieves large improvements:

- **AMBER Discrimination**: Accuracy improves from 67.0 to **76.7** (+9.7 points), F1 from 71.1 to **82.7** (+11.6 points), substantially outperforming the best single strategy (AVISC: 70.7/75.45).
- **POPE-MSCOCO (ALL)**: Accuracy improves from 82.04 to **85.79**, F1 from 80.42 to **83.44**, surpassing ICD (83.26/82.53) and ConVis (83.00) despite being trained only on AMBER data.

On **InstructBLIP-7B**, Octopus also achieves the best performance: AMBER discrimination accuracy 74.00/F1 79.70, and POPE-MSCOCO ALL accuracy 84.79/F1 83.43. These results demonstrate Octopus's advantage in discriminative settings where precise object existence judgments are required.

5.4. Qualitative Analysis

Figure 2 shows the proportion of training samples where each CD strategy is effective on generative and discriminative tasks. Each individual strategy resolves a distinct, non-overlapping subset of hallucinations, and only $\sim$10% of cases are addressed by all three simultaneously. This confirms the necessity of dynamic strategy selection.

Figure 5 presents a token-level qualitative analysis. At different positions within a single generated sentence, the optimal strategy shifts: some tokens benefit from VCD to counteract language priors, others from M3ID to maintain visual grounding, and some from AVISC to correct attention bias. Octopus correctly identifies these per-token needs and selects accordingly.

Figure 6 shows qualitative comparisons of image descriptions generated by the base model, individual CD strategies, and Octopus. While VCD, M3ID, and AVISC each reduce certain hallucinations, they may introduce errors on tokens for which they are not suited. Octopus produces the most accurate description by appropriately switching strategies.

Action Selection Statistics. To further understand Octopus's behavior, we analyze the distribution of selected actions across the AMBER generative test set on LLaVA-

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Table 1. Comparison with the state-of-the-art methods for the generative task across three datasets. For reference, we also provide the results of fine-tuning LVLMs. † signifies results reproduced with the official implementation codes.

LVLMs	Method	MMHalBench		Object HalBench		AMBER			
		Score ↑	HalRate ↓	CHAIR_s ↓	CHAIR_i ↓	CHAIR ↓	Cover. ↑	HalRate ↓	Cog. ↓
Referenced Results (Not Directly Comparable)									
GPT-4V [1]	Base	3.49	0.28	13.6	7.3	4.6	67.1	30.7	2.6
LLaVA-v1.5 -7B[18]	HACL [11]	2.13	0.50	-	-	-	-	-	-
	POVID [39]	2.08	0.56	48.1	24.4	-	-	-	-
	EOS [37]	2.03	0.59	40.3	17.8	5.1	49.1	22.7	2.0
	HA-DPO [38]	1.97	0.60	39.9	19.9	6.7	49.8	30.9	3.3
	HALVA [26]	2.25	0.54	-	-	6.6	53.0	32.2	3.4
Comparable Results									
LLaVA-v1.5 -7B [18]	Base	1.59	0.72	25.0	9.2	8.0	44.5	31.0	2.2
	+ LCD [20]	-	-	61.0	16.1	-	-	-	-
	+ ICD [33]	-	-	47.4	13.9	-	-	-	-
	+ OPERA [9]	2.15	0.54	45.1	22.3	-	-	-	-
	+ VCD [14]	1.96†	0.64†	23.6†	8.4†	6.7	46.5	27.8	2.0
	+ M3ID [8]	2.14†	0.61†	23.2†	7.3†	6.0	48.9	26.0	1.5
	+ AVISC [35]	2.19†	0.59†	22.1†	7.8†	6.3	46.6	25.6	2.0
	+ Octopus (Ours)	**2.61**	**0.50**	**20.8**	**6.6**	**4.8**	**49.2**	**23.4**	**1.2**
Instruct-BLIP [7]	Base	1.84	0.64	0.7	9.1	8.4	46.4	31.1	2.6
	+ LCD [20]	-	-	17.4	10.7	-	-	-	-
	+ ICD [33]	-	-	15.2	8.0	-	-	-	-
	+ OPERA [9]	-	-	16.6	6.8	-	-	-	-
	+ VCD [14]	1.75†	0.64†	0.8†	8.9†	7.6	47.7	29.9	2.2
	+ M3ID [8]	1.70†	0.65†	0.9†	7.6†	6.9	47.2	27.5	2.2
	+ AVISC [35]	2.03†	0.59†	0.7†	8.3†	6.7	46.7	28.0	2.6
	Octopus (Ours)	**2.31**	**0.49**	**0.5**	**6.8**	**6.1**	**48.5**	**22.2**	**1.3**

Table 2. Comparison with the state-of-the-art methods for the discriminative tasks across two datasets. † signifies results reproduced with the official implementation codes.

	AMBER		POPE_MSCOCO							
	Discrimination		Random		Popular		Adversarial		ALL	
	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1	Accuracy	F1
LLaVA-1.5-7B[18]	67.00	71.10	83.77	81.94	82.57	80.86	79.77	78.47	82.04	80.42
+ICD [33]	-	-	**87.51**	83.28	83.15	83.91	79.13	80.41	83.26	82.53
+ConVis [22]	-	-	84.70	-	83.20	-	81.10	-	83.00	-
+OPERA [9]	-	-	84.40	-	83.40	-	81.20	-	83.00	-
+VCD [14]	67.30	71.10	85.43	83.99	83.17	81.94	80.27	79.49	82.96	81.81
+M3ID [8] †	67.25	70.90	86.13	81.85	82.07	80.77	79.50	78.15	82.57	80.26
+AVISC [35]	70.70	75.45	84.67	82.21	83.67	81.27	81.83	79.55	83.39	81.01
+Octopus (Ours)	**76.70**	**82.70**	**87.51**	**85.40**	**85.20**	**84.19**	**82.22**	**81.44**	**85.79**	**83.44**
InstructBLIP [7]	68.20	74.60	81.53	81.19	78.47	78.75	77.43	78.00	79.14	79.31
+ICD [33]	-	-	84.36	83.82	77.88	78.70	75.17	77.23	79.14	79.92
+OPERA [9]	-	-	84.57	83.74	78.24	79.15	74.59	76.33	79.13	79.74
+VCD [14]	69.65	75.90	82.03	81.56	79.13	79.20	77.23	77.72	79.46	79.49
+M3ID [8] †	69.05	75.25	82.33	81.53	80.90	80.42	78.53	78.49	80.59	80.15
+AVISC [35]	72.60	78.60	86.03	84.41	84.27	82.77	81.83	80.67	84.04	82.62
+Octopus (Ours)	**74.00**	**79.70**	**86.63**	**85.30**	**84.90**	**83.55**	**82.83**	**81.43**	**84.79**	**83.43**

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Table 3. **Ablation study.** We select different numbers of contrastive decoding methods as candidates to demonstrate the effectiveness of our approach. "Str1, Str2, Str3" indicate CD strategy VCD [14], M3ID [8] and AVISC [35] in Sec. 4.2, respectively.

	Str1	Str2	Str3	Octopus	CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1					8.0	44.5	31.0	2.2
2	✓	✓	✓		6.9	46.2	26.1	2.2
3	✓	✓		✓	5.5	48.7	25.8	1.5
4	✓		✓	✓	5.7	48.2	25.3	1.5
5		✓	✓	✓	5.5	48.4	26.2	1.6
6	✓	✓	✓	✓	**4.8**	**49.2**	**23.4**	**1.2**

1.5-7B. The Null action is most frequent (∼52% of tokens), confirming that only a minority of generation steps are hallucination-prone. Among CD strategies, AVISC is selected most often (∼24%), followed by VCD (∼14%) and M3ID (∼10%). This distribution aligns with the prevalence of attention misalignment as the dominant hallucination cause in AMBER captions, while language prior bias and fading visual grounding are secondary but non-negligible contributors. These statistics further validate the design choice of including a Null action: forcing a CD adjustment at every step would degrade fluency on non-hallucinated tokens.

5.5. Ablation Study

Table 3 ablates the effectiveness of combining CD strategies within Octopus.

Row 1 (base model) achieves CHAIR 8.0. Row 2 shows that naively applying all three strategies simultaneously (without Octopus) yields CHAIR 6.9, a modest improvement. Rows 3–5 show that Octopus with any two strategies outperforms either strategy alone (CHAIR 5.5–5.7 vs. individual strategies). Row 6 (full Octopus with all three strategies) achieves **CHAIR 4.8**, demonstrating that more candidate strategies further benefit the dynamic selection.

These results confirm that: (1) fixed application of a single or combination of strategies is inferior to learned dynamic selection; (2) Octopus benefits from a richer strategy pool; (3) the decision mechanism is the key contributor to performance gains.

5.6. Design Analysis

Table 4 analyzes design choices for the Octopus module.

Reward signal (rows 1–2). We compare using Coverage as the sole ranking criterion versus averaging multiple metrics. Average scoring (CHAIR 5.0) outperforms Coverage-only (CHAIR 5.4), suggesting that multi-metric rewards provide better guidance.

Training algorithm (rows 3–4). Monte Carlo [34] tree search sampling achieves CHAIR 5.1, comparable to our method, but is computationally heavier. PPO [27] yields CHAIR 5.8 and is less stable, confirming that DPO is a better fit for our offline preference setting.

Table 4. The effects of different settings for our method.

		CHAIR ↓	Cover. ↑	Hal. ↓	Cog. ↓
1	Cover	5.4	**50.1**	26.1	1.2
2	Average	5.0	49.8	23.0	1.4
3	Monte Carlo [34]	5.1	48.1	23.9	1.3
4	PPO [27]	5.8	47.5	**23.0**	1.5
5	Pooling [3]	6.2	44.8	25.0	1.6
6	Cross-atteition [30]	5.1	49.0	24.2	1.3
7	Ours	**4.8**	49.2	23.4	**1.2**

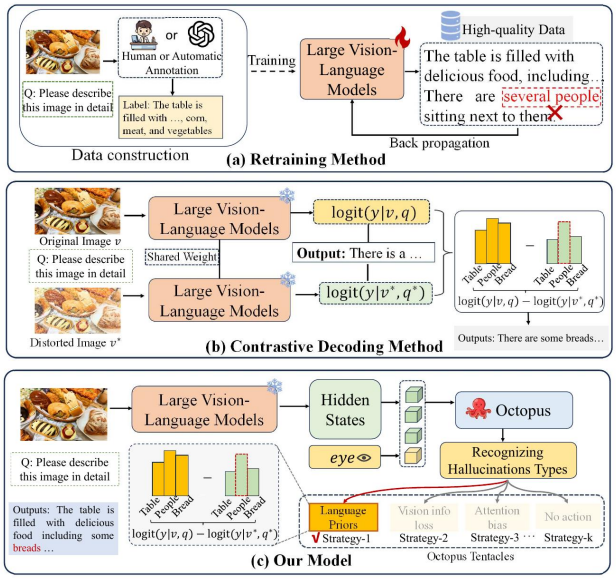

Figure 1. **Paradigm comparison of different hallucination alleviation methods.** Training-based methods fine-tune the LVLM weights; CD-based methods apply a single fixed strategy; Octopus dynamically selects among multiple strategies per token without modifying the LVLM.

Architecture for aggregation (rows 5–6). Pooling [3] of hidden states (CHAIR 6.2) is significantly worse than our cls-token approach. Cross-attention [30] achieves CHAIR 5.1 but is more complex. Our final design (row 7) with a prepended cls token and transformer encoder achieves the best CHAIR 4.8, demonstrating the importance of an expressive yet efficient aggregation mechanism.

6. Conclusion

We have presented **Octopus**, a dynamic contrastive decoding framework that alleviates hallucinations in LVLMs by learning to select, at each token generation step, the

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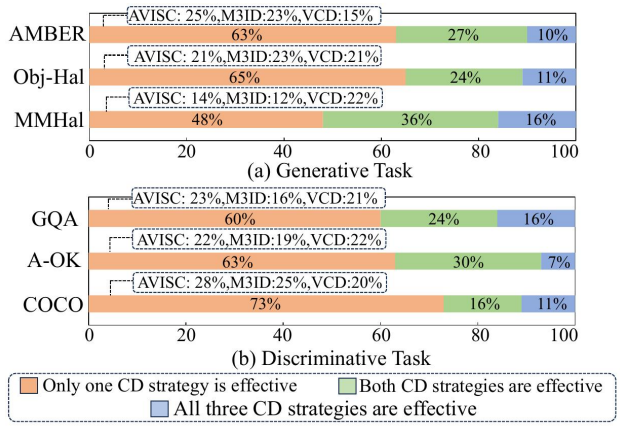

Figure 2. **Proportion of effective samples** using different CD methods for (a) Generative Task and (b) Discriminative Task. Each strategy is only effective on a subset of samples, motivating dynamic selection.

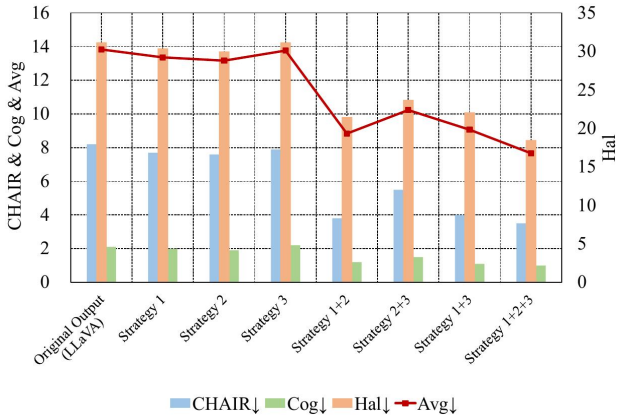

Figure 3. **Token-level hallucination quantitative evaluation.** Different tokens within the same generated sequence require different CD strategies for optimal hallucination suppression.

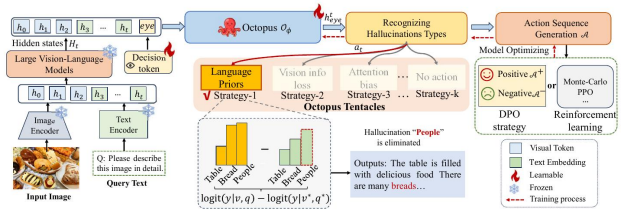

Figure 4. **Overview of our Octopus method.** At each decoding step, the decision token (eye) attends to the LVLM's hidden states via a small transformer encoder and selects the appropriate CD strategy (tentacle) from {VCD, M3ID, AVISC, Null}. The LVLM weights are frozen; only the Octopus module is trained.

most appropriate CD strategy from a candidate pool. Our key insight is that hallucination causes are hybrid—varying

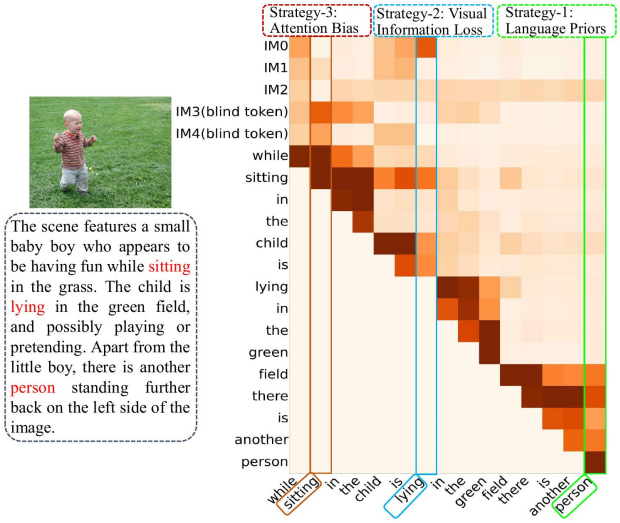

Figure 5. **Token-level hallucination qualitative analysis.** Visualization of which CD strategy is selected at each token position, showing that the dominant hallucination cause varies within a single generated sequence.

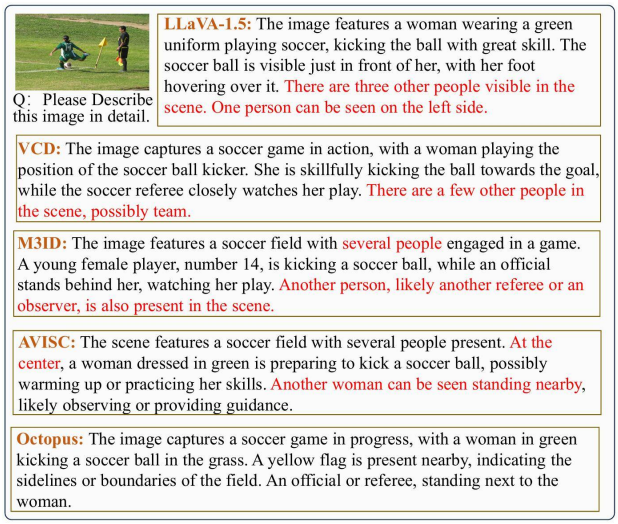

Figure 6. **Qualitative comparison** of generated descriptions with different CD strategies and Octopus. Octopus produces the most accurate description by dynamically selecting the optimal strategy per token.

across samples and across tokens within a single sequence—making any fixed single-strategy approach inherently suboptimal. Octopus addresses this by introducing a lightweight transformer module with a learnable decision token trained via DPO on hallucination-metric-derived preference pairs, all without modifying the LVLM weights. Extensive experiments on five benchmarks with two LVLM families demonstrate that Octopus achieves state-of-the-art

hallucination reduction while remaining plug-and-play and easily extensible. We believe the dynamic per-token strategy selection paradigm opens promising avenues for future research in multimodal reliability and trustworthy AI. Promising future directions include extending Octopus to video-language models where temporal grounding introduces additional hallucination modes, and exploring joint finetuning of the decision module with the LVLM backbone to further tighten the synergy between strategy selection and generation.

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